FEU INSTITUTE OF TECHNOLOGY

COLLEGE OF COMPUTER STUDIES AND MULTIMEDIA ARTS

CS0065
(INTELLIGENT SYSTEMS)

TECHNICAL ASSESSMENT

2

RULE – BASED AGENT

Student Name	Mar Vincent G. Inoy
Section:	AN41
Professor:	Ms. Crisola Tan
Date:	

I. BACKGROUND INFORMATION

A Rule-Based Agent is a type of intelligent agent that operates by following predefined rules to decide its actions. In this lab, students will work with a vacuum cleaner agent that can clean rooms or move between rooms depending on their state (dirty or clean).

The goal is to understand how agents perceive the environment, act according to rules, and update the environment.

II. TECHNICAL ASSESSMENT

RULE – BASED AGENT

Objective

At the end of this laboratory activity, students are expected to:

1. Explain the concept of rule-based agents.
2. Implement a rule-based agent in Python.
3. Interact with an environment by perceiving and acting based on rules.
4. Observe and analyze the behavior of an agent in a controlled environment.

Materials Needed

- Computer with Python 3.x installed
- Text editor or IDE (e.g., VSCode, PyCharm)
- Optional: Jupyter Notebook for interactive testing

General Instructions

1. This **Technical Assessment (TA4)** must be completed individually.
2. Follow the Task below.
3. Include clear comments in your code to describe its logic.
4. Each program should display the correct input, process, and output.
5. Ensure that your script displays clear and readable outputs.
6. Compile all Python codes and corresponding outputs into one PDF file.
7. Save it as **TA2_lastname.pdf**
   (Example: TA2_Patilano.pdf)
8. Submit the **python file** and **PDF** file.

Laboratory Task

Pre-Lab Preparation

1. Review the Python concepts:
   - Classes and objects
   - If-Else statements

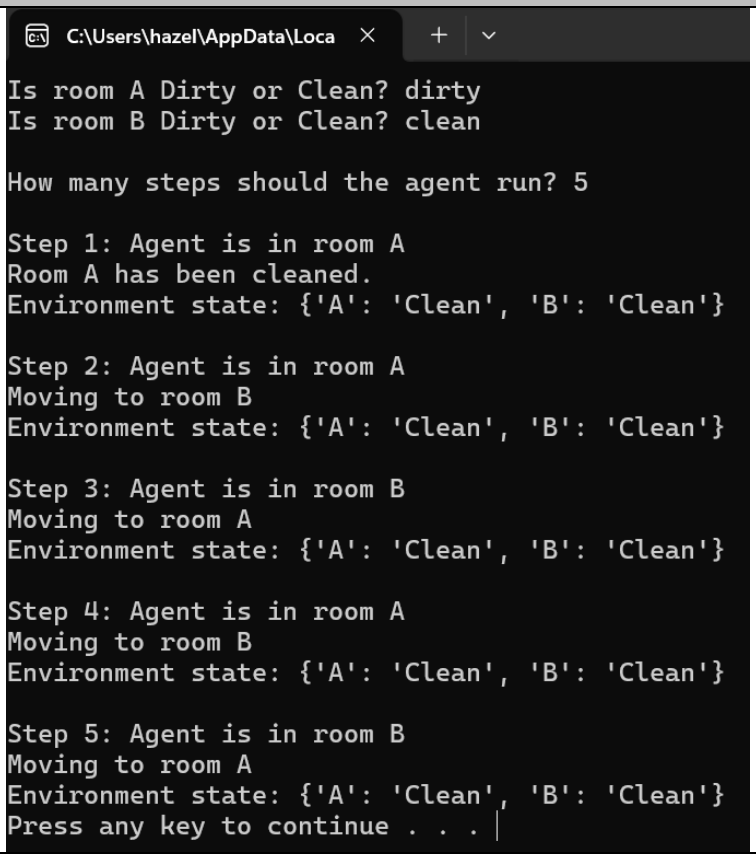
- Loops (for loop)
- Functions / methods

2. Understand the environment:
   - Two rooms: A and B
   - Each room can be Dirty or Clean

Task 1: Set Up the Environment

1. Initialize two rooms using a Python dictionary.
2. Allow the user to set the **initial state** of each room (dirty or clean).
3. Write a function to **check if a room is dirty** and a function to **clean a room**.

Sample Output



Task 2: Implement the Rule-Based Agent

1. Create a class for the agent.
2. Implement the **perceive-and-act method**:
   - Rule 1: If the current room is dirty, clean it.
   - Rule 2: If the current room is clean, move to the other room.
3. Implement a **move method** to switch rooms.

Task 3: Run the Simulation

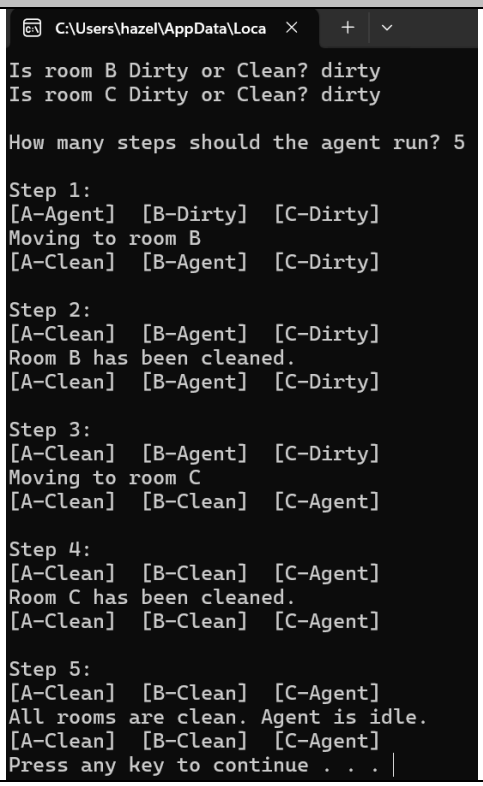
1. Ask the user how many steps the simulation should run.
2. Run the agent in a loop for the specified steps.

3. After each step, print:
    - The agent's current location
    - The action taken (cleaned/moved)
    - Current state of the environment

Optional (Bonus Task)

- Add a **third room (C)** and update the agent logic.
- Make the agent **randomly choose which dirty room to clean next**.
- Visualize the agent moving between rooms using a **text-based grid**.

Sample Output (Bonus Task)



Your Code - Laboratory Tasks ⇩

```
def is_dirty(environment, room):
    """Return True when the selected room is dirty."""
    return environment[room] == "Dirty"

def clean_room(environment, room):
    """Change the selected room's state to Clean."""
    environment[room] = "Clean"
```

```
class VacuumAgent:
    """Rule-based agent that operates in rooms A and B."""

    def __init__(self, environment, starting_room="A"):
        self.environment = environment
        self.current_room = starting_room

    def move(self):
        """Move to the other room."""
        self.current_room = "B" if self.current_room == "A" else "A"

    def perceive_and_act(self):
        """Agents Rules

        Rule 1: If the current room is dirty, clean it.
        Rule 2: If the current room is clean, move to the other room.
        """
        if is_dirty(self.environment, self.current_room):
            clean_room(self.environment, self.current_room)
            return f"Cleaned room {self.current_room}."

        previous_room = self.current_room
        self.move()
        return f"Moved from room {previous_room} to room {self.current_room}."

def read_room_state(room):
    """Validate initial state"""
    while True:
        state = input(f"Enter the initial state of room {room} (dirty/clean): ").strip().lower()
        if state in {"dirty", "clean"}:
            return state.capitalize()
        print("Invalid input. Please enter dirty or clean.")

def read_step_count():
    """Validate simulation of steps"""
    while True:
        try:
            steps = int(input("How many simulation steps should run? ").strip())
            if steps >= 1:
                return steps
        except ValueError:
            pass
        print("Invalid input. Enter a whole number greater than or equal to 1.")
```

```
def format_environment(environment):
    """Return to readable form"""
    return f"A={environment['A']}, B={environment['B']}"

def main():
    """Simulation and Display"""
    print("Rule-Based Vacuum Cleaner Agent")
    print("--------------------------------")

    # The environment is represented by a dictionary with two rooms.
    environment = {
        "A": read_room_state("A"),
        "B": read_room_state("B"),
    }
    steps = read_step_count()
    agent = VacuumAgent(environment)

    print(f"\nInitial environment: {format_environment(environment)}")
    print(f"Starting location: room {agent.current_room}")

    for step in range(1, steps + 1):
        action = agent.perceive_and_act()
        print(f"\nStep {step}")
        print(f"Current location: room {agent.current_room}")
        print(f"Action taken: {action}")
        print(f"Environment: {format_environment(environment)}")

    print("\nSimulation complete.")

if __name__ == "__main__":
    main()
```

Task 4: Analysis Questions

After running the simulation, answer the following:

1. What rules does the agent follow?
2. How does the agent decide when to clean and when to move?
3. What happens if both rooms are already clean at the start?
4. How would you modify the agent to handle **more than two rooms**?

Answer the Questions ⇩

1. First one is cleaning the room if it is dirty then cleans it. Second is move to another room if it is already cleaned.

2. The agents checks if it is is_dirty then cleans it if it is true. If it is false it calls move and changes to the other room.

3. It doesn't have anything to clean so what will happen is it will just go switch rooms in every steps.

4. I think I might not use the names room a and room b. Instead, I will make the rooms names into an ordered list. Then make the agent just scan for dirty rooms then go to it.